

Exam Date & Time: 15-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Pattern classification [IT_5006]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A university plans to replace traditional ID cards with a biometric-based attendance system that uses both facial recognition and fingerprint verification. During implementation, the system faces challenges in recognizing students under varying lighting conditions and partial fingerprint captures. (5)
- A) Illustrate the challenges this system faces from the perspective of pattern recognition design. Explain how feature extraction, classification, and decision-making steps play a role here. Also, classify the type of pattern recognition method (statistical, structural, or syntactic) primarily used and justify your answer
- B) A retail company collects large amounts of purchase history and demographic data from its customers but lacks labeled data. The company aims to group customers with similar shopping behaviors to design targeted marketing strategies. (5)
- Describe how unsupervised learning techniques can be applied in this scenario. Explain the concept of feature space and how clusters represent implicit classes. Discuss how the resulting decision boundaries differ from those in supervised learning and the challenges involved in evaluating cluster quality.
- 2) An AI company is building two products: (5)
- Product A: Automatically reads and digitizes handwritten documents.
 - Product B: Converts spoken language into text commands for smart assistants.
- A) Both systems rely on pattern recognition but process different types of data (visual and auditory). (5)
- Compare and contrast the pattern recognition processes in Product A and Product B. Discuss how data representation, feature selection, and classification techniques differ between the two. Further, identify the type of pattern (statistical, structural, or syntactic) each system most likely employs and justify with appropriate examples.
- B) A team of researchers is developing an autonomous drone navigation system that learns optimal flight paths through trial and error. The drone receives rewards for efficient routes and penalties for collisions or deviations from the target zone. (5)

Analyze how reinforcement learning serves as a paradigm of pattern recognition in this

scenario. Explain how the drone's state, action, and reward can be represented in feature space, and discuss how decision boundaries evolve dynamically as learning progresses. Compare this paradigm with supervised and unsupervised learning in terms of feedback and data requirements.

- 3) A human-computer interaction (HCI) team is developing a gesture recognition system to classify hand movements into actions like swipe left, swipe right, zoom in, and zoom out.
 - A) Analyze how the distance metric selection (e.g., Euclidean vs. Manhattan) influences the classifier's sensitivity to noisy sensor data. (5)
Compare the effectiveness of 1-NN, k-NN, and weighted k-NN in handling ambiguous gestures.
 - B) A computer vision team is building an image classification model to identify plant species from leaf images. Initially, the training dataset contains over 50,000 labeled samples, leading to high computation time in the k-Nearest Neighbor classifier. To reduce processing time without losing accuracy, the team applies prototype selection techniques such as Condensed Nearest Neighbour (CNN) and Edited Nearest Neighbour (ENN). (5)

Explain how CNN and ENN algorithms can be applied to optimize the training set in this scenario. Compare their working principles, objectives, and effects on classification accuracy and computational efficiency. Illustrate how each method modifies the prototype set and why such reduction is beneficial.
- 4) A hospital's AI department is developing a diagnostic system that classifies patients as either *disease-positive* or *disease-negative* based on test results (e.g., blood sugar, cholesterol, BMI). Each class has known prior probabilities and likelihoods estimated from historical data.
 - A) Illustrate how a Bayes classifier can be used to make optimal decisions in this medical diagnosis scenario. Using the concept of minimum error rate classification, explain how the decision boundary between the two classes is determined. Discuss how changing prior probabilities (e.g., due to prevalence of disease) affects the classification outcome and error rate. (5)
 - B) A financial institution uses customer data (income, credit history, debt ratio, employment type) to classify applicants as *low-risk* or *high-risk* for loan approval. The data contains overlapping cases where some high-risk customers have similar profiles to low-risk ones, making perfect separation impossible. (5)

Analyze how an SVM classifier with slack variables can handle this real-world dataset where classes are not perfectly separable. Explain the trade-off between margin width and classification error, and how the regularization parameter (C) influences this balance.
- 5) A tech company is developing an automated document recognition system that converts scanned documents into editable text using Optical Character Recognition (OCR). To improve recognition accuracy across different fonts and languages, the developers combine multiple classifiers using bagging, boosting, and stacking techniques. (5)

Analyze how ensemble learning methods (bagging, boosting, and stacking) can improve

the performance of OCR systems in this scenario. Explain the working principles of each technique and how they address errors from individual classifiers. Discuss the trade-offs between computational complexity and accuracy when combining multiple classifiers in document recognition tasks

- B) A digital library project aims to organize scanned research papers into categories (e.g., computer science, biology, economics). Each document contains a mix of text blocks, figures, tables, and equations. The team decides to use feature extraction techniques to represent the visual and textual structure of documents before classification. Explain how feature extraction in document processing can be used to classify documents in this scenario. Identify the types of features (e.g., text density, font size, object layout, keyword frequency) that could help in distinguishing document categories. Discuss how extracted features can be represented numerically and how they contribute to pattern recognition or machine learning models for document classification. (5)

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Event Name		Subject	
058-END SEMSTER-ODD-2025		IT_5006:Pattern classification	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name		Subject	
058-END SEMSTER-ODD-2025		IT_5006:Pattern classification	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	25(50%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)

Total	10	50 (100%)
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